A Double Pulse Overlapping Laser Diode Driver With Minimum 100-ps Pulse for LiDAR System

Zhiguo Tong, *Student Member, IEEE*, Junwei Huang, *Student Member, IEEE*, Xiangyu Mao, *Member, IEEE*, Rui P. Martins, *Life Fellow, IEEE*, and Yan Lu, *Senior Member, IEEE*

***Abstract*— Light detection and ranging (LiDAR) technology is widely spreading to many applications. Among the various ranging methods, direct time of flight (dToF) stands out as particularly promising for real-time and long-range applications. However, the detection resolution and range of dToF LiDAR systems depend on the minimum width and maximum power of the emitted laser pulse. This work introduces a new laser diode driver (LDD) structure for high-resolution and long-range LiDAR transmitter (TX) system. It features a double pulse overlapping (DPO) scheme to generate ultra-short laser pulses spanning from 100 ps to 2 ns, achieving output pulsewidths significantly shorter than the control signal. Furthermore, two package configurations are implemented and an interleaved bonding method is analyzed and utilized to reduce the parasitic inductance and hence to improve the peak optical power. The parasitic effect analysis and system implementation circuit techniques are also presented. Fabricated in a 180-nm high-voltage BCD process, the two implemented TX systems achieve a minimum pulsewidth of 100 ps or a maximum output current of 15 A within 1-ns pulsewidth.**

***Index Terms*— Diode on chip, diode on interposer, diode on PCB, direct time of flight (dToF), double pulse overlapping (DPO), high-resolution light detection and ranging (LiDAR), laser diode driver (LDD), long-range LiDAR, transmitter (TX).**

I. Introduction

LIGHT detection and ranging (LiDAR) technology has a wide utilization in the reconstruction of 3-D maps for various applications such as augmented reality (AR), virtual reality (VR), as well as in autonomous vehicles. In the past decades, different methods for 3-D ranging have been proposed, including stereo vision, projection with structured light, and time of flight (ToF) [1], [2], [3]. Among these ranging methods, data extraction of ToF is much easier, leading to lower power consumption and faster distance calculation. Therefore, ToF stands out as a preferred option for applications requiring real-time capabilities. Generally, ToF techniques can be divided into two categories: indirect ToF (iToF) and direct ToF (dToF). iToF, also known as continuous-wave CW-LiDAR, measures the phase difference between the light source and the detected backscattered light signal. Although the iToF has advantages in high resolution, its application is restricted to short-range scenarios (<10 m) [4]. On the other hand, as illustrated in Fig. 1, dToF, also known as pulsed-LiDAR, relies on the measurement of the round-trip travel time Δt of an optical pulse emitted by the laser diode and reflected from the objects. By using the ToF information, we can calculate the measured distance D as

$$D = c \cdot \Delta t/2 \tag{1}$$

where c is the speed of light (30 cm/ns). The dToF can achieve a long-range detection (>100 m), as theoretically the maximum distance is only limited by the optical power [4], [5].

However, in practical scenarios, the non-ideal characteristics of the emitted light pulses significantly impact the system's performance. As shown in Fig. 2(a), the received optical power P_{RX} will be much lower than that of the transmitted optical power P_{TX} due to the energy transmission loss. For a linear detector scenario, the signal-to-noise ratio (SNR) in the electrical domain is determined by the input optical power P_{RX} and the total input referred noise [6]. Then, leading to

$$\text{SNR} \propto P_{RX} \propto \frac{\alpha \cdot P_{TX}}{D^2} \tag{2}$$

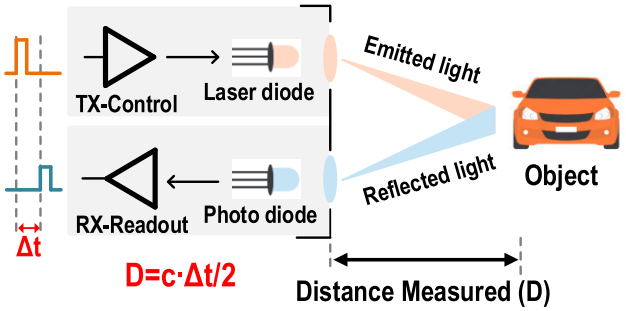


Fig. 1. dToF LiDAR operation principle.

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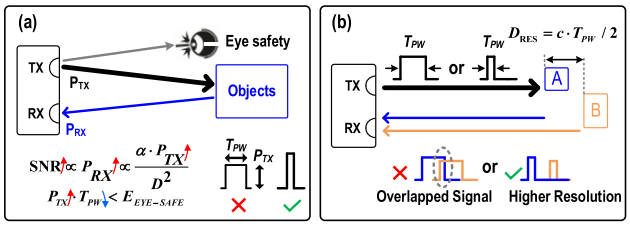


Fig. 2. Challenges of the LiDAR TX system. (a) SNR and eye safety. (b) Detection resolution.

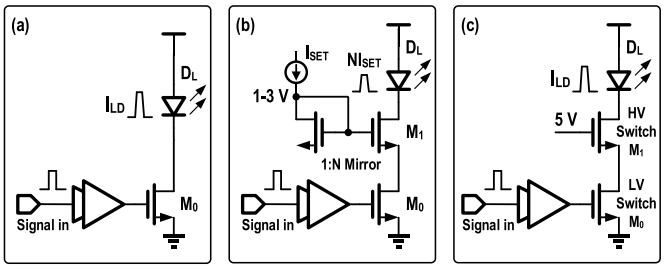


Fig. 3. Conventional FET-controlled LDDs. (a) Basic LDD. (b) Current mirror-based current mode LDD. (c) Cascode structure-based switch mode LDD.

where α is the target reflectivity and D is the measured distance. This equation implies that the SNR drops quadratically with the distance and linearly with the reflectivity. Therefore, the SNR can only be improved by maximizing the transmit power P_{TX} for long-range and low-reflectivity targets.

However, the improvement of P_{TX} leads to eye-safety problems. Eye-safety regulation requires the limitation of the emitted laser energy to the maximum eye-safety energy level, $E_{\mathrm{EYE\text{-}SAFE}}$ [6], [7], [8]. This means that for a transmitted laser pulse with pulsewidth of T_{PW}, we will have

$$P_{\mathrm{TX}} \cdot T_{\mathrm{PW}} < E_{\mathrm{EYE\text{-}SAFE}}. \tag{3}$$

Hence, the increase in P_{TX} requires a shorter pulsewidth T_{PW} to meet the eye-safety regulation.

Moreover, the pulsewidth T_{PW} also determines the detecting resolution D_{RES} [9], [10], [11], [12], [13], with

$$D_{\mathrm{RES}} = c \cdot T_{\mathrm{PW}}/2. \tag{4}$$

As shown in Fig. 2(b), it would be hard to distinguish a distance smaller than D_{RES} due to the overlapping of reflected signals of different objects. According to (4), for applications that require D_{RES} in the order of centimeters, the required T_{PW} is around 100–300 ps. Besides, in LiDARs employing single-photon detection, sub-nanosecond pulses prove beneficial as they match more closely with the speed of the detector.

In conclusion, in order to enhance the performance of dToF LiDAR, emitting a high-peak-power laser pulse with an ultra-short width is crucial for the transmitter (TX) module. In this work, a LiDAR TX module that employs the double pulse overlapping (DPO) scheme to generate ultra-short laser pulses ranging from 100 ps to 2 ns is introduced. Furthermore, we implement two packaging solutions to minimize parasitic inductance, thereby enhancing peak optical power within the short pulsewidth.

The organization of the remainder of this article is the following. Section II presents the conventional and proposed laser diode drivers (LDDs) and provides the detailed analysis of the proposed structure. Section III gives the circuit design and package implementation of the proposed TX module. Besides, an interleaved bonding method is also presented to mitigate the influence of the bonding wires. Section IV exhibits the measurement results of the received optical signal, optical average power, and the comparison with prior works. Finally, Section V draws the conclusions of this work.

II. Proposed LDD

A. Conventional FET-Controlled LDD and Limitation

There are conventionally two kinds of LDD: the FET-controlled LDD and the capacitive discharge LDD (also known as resonant LDD). Fig. 3(a) shows the basic FET-controlled LDD [14], [15]. It uses only one transistor to turn on and off the current path of the laser diode. In other words, both the rising and falling edges of laser diode current should be controlled. This approach ensures that the generated current pulsewidth is the same as the input control signal pulsewidth and it simplifies the generation of the tunable pulsewidth, which is very helpful for signal processing [13]. Besides, some other modified FET-controlled LDD structures emerged in recent years. Fig. 3(b) presents a current mirror-based current mode LDD [16], [17]. It sets the peak current of the laser diode by the current mirror structure. However, the size of M_1 should be large enough to carry the desired high current amplitude as M_1 operates in the saturation region. Fig. 3(c) presents another cascode structure-based switch mode LDD [17], [18], [19]. Instead of using current mirror, the cascode structure consists of a high voltage switch M_1 to carry high voltage and a low voltage switch M_0 as the primary switch. In this way, the power switch M_0 can switch faster because of the low voltage switch used.

TABLE I
Derated Minimum Input Pulsewidth of Pre-Drivers

Manufacture	Part number	Minimum input pulse width
TI	LMG1020	1ns
uPI SEMI	uP1966E	5ns
Infineon	1EDN751x	6ns

While the aforementioned structures resolved certain issues, they all encountered a significant limitation: the generated laser pulsewidth is constrained by the minimum input pulsewidth of the pre-drivers. In Table I, we list a few state-of-the-art commercial pre-drivers with derated minimum input width, where they all exhibit an input timing value exceeding 1 ns. The reason why pre-drivers have the derated minimum input pulsewidth is shown in Fig. 4. As is known to us, when the signal width is close to the sum of rising and falling edge widths of the inverter chain, the signal would gradually attenuate until it completely diminishes. With commonly used 5-V devices in the 180-nm process, the rising and falling edge widths in pre-drivers typically range from 150 to 250 ps [20], [21]. Consequently, it becomes challenging for the pre-drivers to effectively transmit sub-500-ps pulse signal, particularly when considering the process, voltage, and temperature (PVT) variations.

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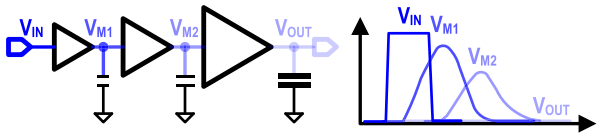


Fig. 4. Simulated waveforms of pre-drivers when the signal width is close to the sum of the rising or falling edge widths.

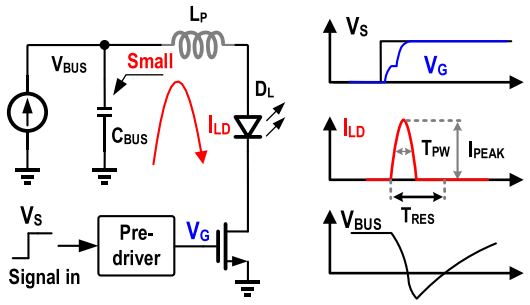


Fig. 5. Conventional capacitive discharge LDD and its operation principle.

TABLE II
CALCULATED V_{BUS} VOLTAGE TO MAINTAIN THE SAME CURRENT

I_{PEAK}(A)	L_p (nH)	t_p (ns)	C_{BUS} (nF)	V_{BUS} (V)
20	1	3	2.05	15
20	1	1	0.23	43
20	1	0.2	0.01	210

B. Conventional Capacitive Discharge LDD and Limitation

Fig. 5 illustrates the structure and operation principle of the conventional capacitive discharge LDD [22], [23], [24]. It utilizes the transistor to just turn on the current path. Subsequently, owing to the use of small capacitor C_{BUS}, the laser diode D_L, the parasitic inductance L_P, and C_{BUS} form a resistor–inductor–capacitor (*RLC*) resonant network in which C_{BUS} discharges and rings up the laser diode current I_{LD}. Consequently, this structure has a fixed full-width at half-maximum (FWHM) T_{PW} determined by the *RLC* network time constant T_{RES}

$$\begin{cases} T_{RES} = 2\pi\sqrt{L_P C_{BUS}} \\ T_{PW} = T_{RES}/3 \\ I_{PEAK} = (V_{BUS} - V_{Diode})\sqrt{C_{BUS}/L_P}. \end{cases} \quad (5)$$

By following this approach, the generated pulsewidth can be shorter than the input pulse of pre-drivers. Hence, the limitation of the minimum input pulsewidth is eliminated.

However, to generate a short current pulse, C_{BUS} needs to be sufficiently small, leading to the decrease of energy stored in C_{BUS}. Consequently, the voltage must be raised significantly to maintain the desired current amplitude [9], [25]. Here, a calculated value in Table II is presented to guarantee a 20-A current pulse. Notably, achieving a 200-ps current pulse necessitates the V_{BUS} voltage to be as high as 210 V, presenting numerous challenges in IC and system design [26].

C. Proposed LDD and Operation Principle

In this work, in order to eliminate the input pulsewidth limitation without using high bus voltage, a new LDD structure is proposed, as shown in Fig. 6. This structure stacks two power transistors beneath the laser diode and utilizes their turn-on overlapping time to generate the ultra-short current pulse. Two switches, M_1 and M_2, are both driven by two ground-referenced pre-drivers. Due to the voltage clamp function of M_2, M_1 only needs to withstand 5 V. In this way, we can greatly reduce the requirements for the pre-drivers.

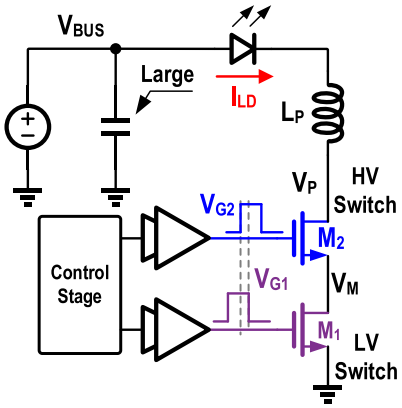


Fig. 6. Proposed LDD structure featured with DPO scheme.

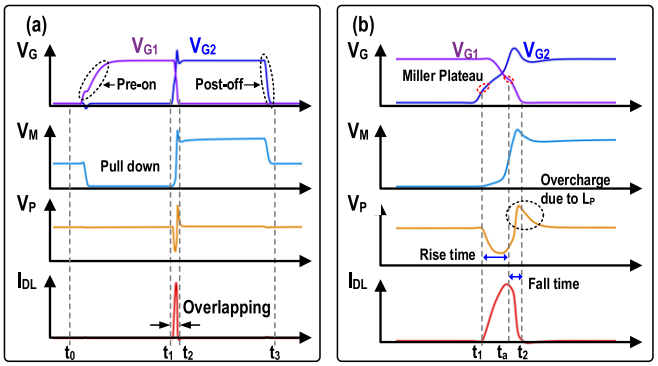


Fig. 7. (a) Operation principle and (b) simulated zoomed-in waveforms from t_1 to t_2.

Fig. 7(a) illustrates the operation principle of the proposed LDD. At $t = t_0$, M_1 turns on gradually to pull down V_M node without generating too much noise. During this period, M_2 remains off, ensuring that no current flows through the laser diode. At $t = t_1$, the turn on of M_2 brings about a rapid pull-down of the V_P node, enabling I_{LD} to rise quickly and emit the desired optical signal. Following this, at $t = t_2$, M_1 turns off to prevent further current flow. Finally, at $t = t_3$, M_2 turns off gradually to conclude this cycle. To have a detailed understanding of the time interval between t_1 and t_2, we present the zoomed-in waveforms in Fig. 7(b).

The laser pulse current initiates from the Miller plateau of V_{G2} and reaches its peak value at the Miller plateau of V_{G1}. Subsequently, the current from the parasitic inductor charges the V_P node, resulting in a rapid decay of the current pulse.

By introducing only one additional power transistor, this structure successfully addresses the limitations of the FET-controlled drivers by transforming the dual-edge control into the single-edge control for each switch without using high bus voltage. Another important advantage is that the duration of the current pulse can be precisely controlled. The overlapping time between the two power transistors determines

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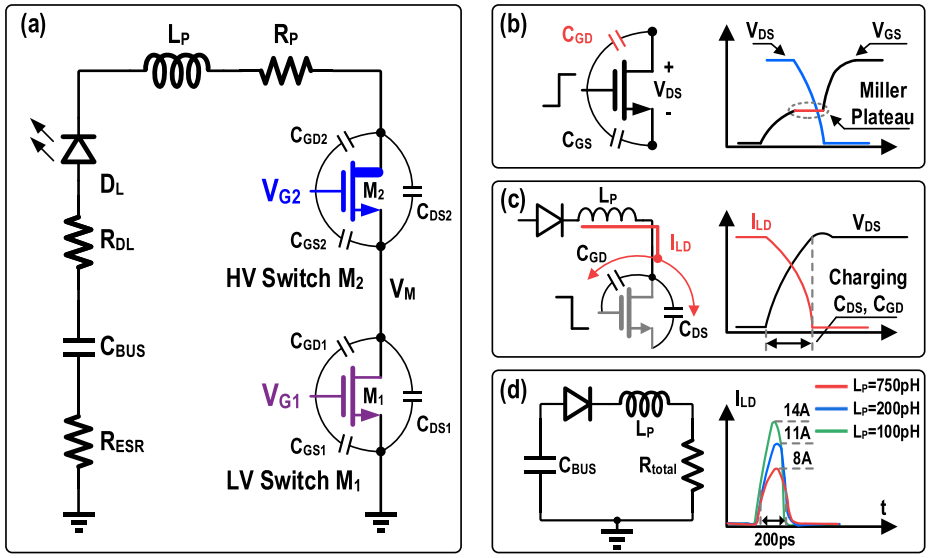


Fig. 8. (a) Detailed circuit in the critical path and three kinds of parasitic effects. (b) Parasitic effect of C_{GS} and C_{GD}. (c) Parasitic effect of L_P, C_{GD}, and C_{DS}. (d) Parasitic effect of L_P.

the pulse duration, which theoretically can be set to any desired value, providing significant flexibility in pulse generation.

D. Parasitic Effect Analysis

Due to the desired laser pulsewidth falling within the range of 100–300 ps, the parasitic components within the circuit notably influence the generated pulsewidth and amplitude. Fig. 8(a) presents the detailed circuit in the critical path, including the parasitic capacitors C_{GS}, C_{GD}, and C_{DS}, the parasitic resistors R_P, R_{DL}, and R_{ESR}, and the parasitic inductor L_P. The parasitic effects of these components can be summarized in three aspects as follows.

First, as illustrated in Fig. 8(b), the capacitances C_{GS} and C_{GD} determine the on/off behavior of the MOSFET. For instance, during the turn-on period, V_{GS} does not immediately rise to 5 V. Instead, it undergoes three necessary steps. Initially, C_{GS} is charged to the threshold voltage V_{TH} to activate the power switch. Then, as V_{DS} decreases, C_{GS} stops being charged, while C_{GD} begins to get charged because of the Miller effect. Therefore, V_{GS} remains constant during this step, creating the so-called Miller plateau [13], [27]. Finally, once V_{DS} reaches zero, V_{GS} begins to rise again until it reaches its final value.

Second, as presented in Fig. 8(c), the falling edge time of the laser diode current I_{LD} is determined by the inductor L_P and capacitors C_{DS} and C_{GD}. After the MOSFET turns off, the current I_{LD} does not instantly drop to zero. Instead, L_P, C_{DS}, and C_{GD} form an LC resonant tank to dissipate the remaining energy in the system. Therefore, it is essential to minimize the parasitic values of L_P, C_{DS}, and C_{GD} to reduce the falling edge time.

Third, as shown in Fig. 8(d), the parasitic inductor L_P and the total resistance R_{TOTAL} that includes all parasitic resistors in the path determine the peak current amplitude of I_{LD}. For a short pulse, we can express its laser diode voltage V_{LD} as

$$V_{LD} = V_{BUS} - I_{LD}R_{TOTAL} - L_P\frac{dI_{LD}}{dt}. \tag{6}$$

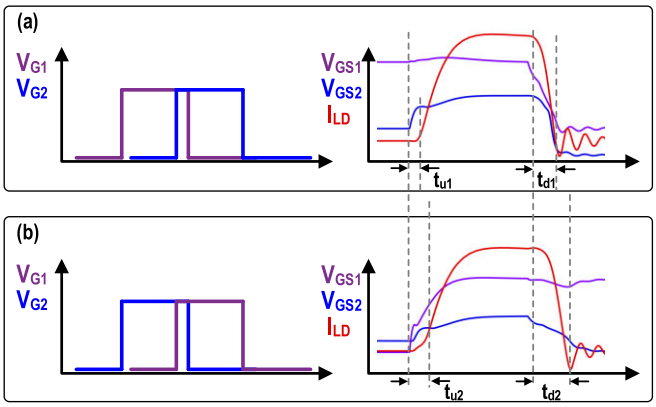


Fig. 9. Two kinds of control sequences. (a) V_{G1} switches on first. (b) V_{G2} switches on first.

The precise relationship between the voltage V_{LD} and current I_{LD} can be obtained by referencing the datasheet of used diode. As a simplification, a higher voltage V_{LD} results in a greater current I_{LD}. Hence, in order to achieve a larger current amplitude, it is necessary to minimize the values of R_{TOTAL} and L_P to a sufficient extent.

E. Comparison of Two Control Sequences

As previously mentioned, the overlapping time of two switches is utilized to control the laser diode current. Therefore, as shown in Fig. 9, there are two kinds of control sequences: the first case where V_{G1} switches on first and the second case where V_{G2} switches on first. Although the steady states under the two control sequences are the same, their transient switching processes are different.

In the first case, as shown in Fig. 9(a), triggering the increase in I_{LD} simply requires the turning on of M_2 with M_1 turned on in advance. On the other hand, triggering the decrease of I_{LD} requires the turning off of the low voltage switch M_1 that has lower parasitic capacitance when compared with the high voltage switch M_2.

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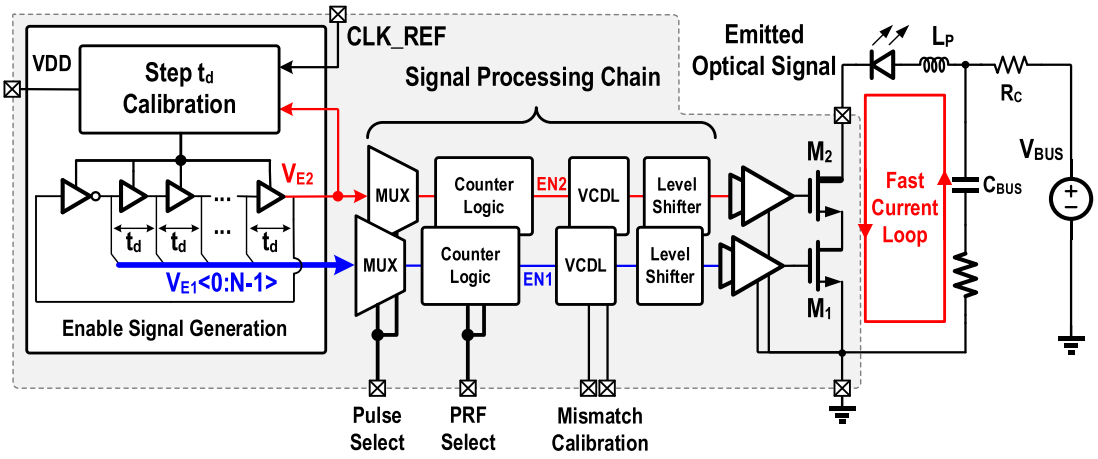


Fig. 10. System block diagram of the proposed LDD.

However, the second case involves a different process. For the current rising edge, the activation of M_1 needs to first pull down the source voltage of M_2. Afterward, with the source voltage sufficiently reduced, M_2 can then be turned on. Thus, in the second case, we must turn on sequentially both M_1 and M_2, one after the other. On the other hand, for the falling edge of the current, high voltage switch M_2 needs to close, which is clearly slower than the turning off of M_1. In conclusion, the first case has a faster switching performance and hence will be adopted in this work.

III. System Architecture and Circuit Implementations

A. Control System Block Diagram

Based on the operational principle of the proposed LDD, the accurate control of two overlapping signals with overlap timings ranging from 100 ps to 2 ns is necessary. This is because the phase difference directly influences the generated laser pulsewidth. However, achieving a control accuracy of 100-ps level and ensuring its robustness across PVT changes presents a significant challenge, especially when using a 0.18-μm HV BCD process. In this work, thanks to the DPO scheme, the generated laser pulsewidth only depends on the relative time of control signal rather than the absolute time.

Fig. 10 presents the system block diagram of the implemented LDD. The circuit in the gray dash highlights the designed integrated pre-driver proposed in this work. It mainly includes three parts: the ring oscillator for signal generation, the signal processing chain, and the power stage (M_1, M_2, and the associated pre-drivers).

Fig. 11 illustrates the simulated pulsewidth variation results of single pulse transmission and double pulse transmission under different PVT cases, including SS, FF, and TT corners, with voltage ranging from 1.6 to 2.0 V and temperature from 27 °C to 80 °C. The pulsewidth variation of the single pulse transmission ranges from 14 to 48 ps, while for double pulse transmission, it is only 1.4–4.3 ps. This difference arises because, for double pulse transmission, we can match the transmission path effectively to mitigate the effects of PVT changes.

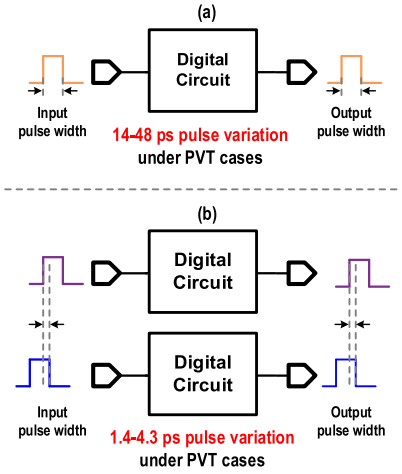


Fig. 11. Pulsewidth variation under different PVT cases of (a) single pulse transmission and (b) double pulse transmission.

Hence, the precise generation of two overlapping pulses with accurate timing and ensuring their matching transmission paths becomes quite important.

In this work, we utilize a ring-oscillator-based signal generation circuit, which includes N delay cells with the same delay t_d and an inverter. Fig. 12 illustrates the working principle and waveforms of the proposed circuits. In this configuration, V_{E2} directly serves as the output enable signal of the power switch M_2. Upon the ring oscillator begins oscillating, each adjacent signal $V_{E1}\langle x\rangle$ and $V_{E1}\langle x+1\rangle$ will share the same step delay of t_d. Consequently, a sequence of enable signals V_{E1} for power switch M_1 are produced by the ring oscillators. Also, the overlap time T_{OL} between $V_{E1}\langle x\rangle$ and V_{E2} can be expressed as

$$T_{\mathrm{OL}} = x \cdot t_d \tag{7}$$

where x ranges from 0 to $N-2$. Besides, to ensure the robustness of step delay t_d, V_{E2} is simultaneously fed back to the calibration unit for comparison with a reference signal $\mathrm{CLK_{REF}}$ of a fixed period T. Then, the power supply voltage can be adjusted to make sure that

$$T = 2N \cdot t_d. \tag{8}$$

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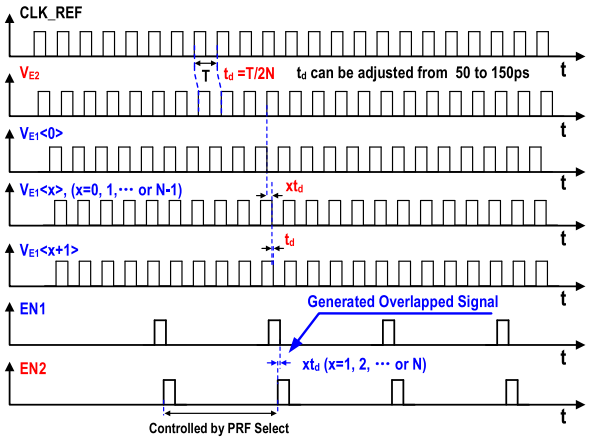


Fig. 12. Working principle and waveforms.

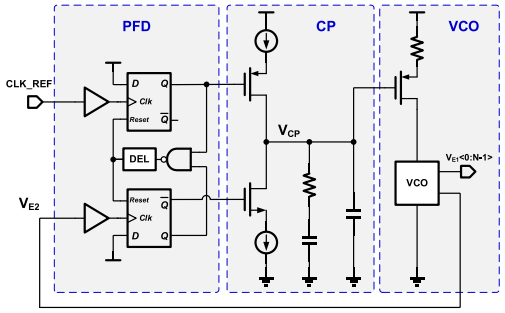


Fig. 13. CP-based PLL for step t_d calibration.

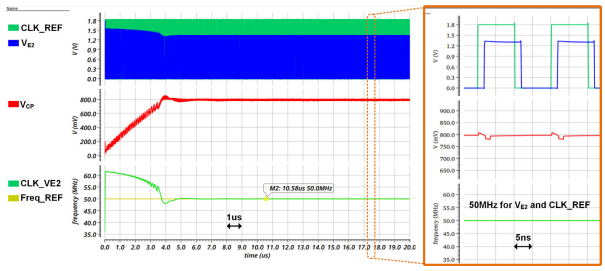


Fig. 14. Simulate results of the CP-based ring oscillator.

With an extended period of $2N$ times, we can realize the step t_d calibration circuit with a low-frequency phase-locked loop (PLL) circuit, as shown in Fig. 13. The PLL circuit utilizes the phase and frequency detector (PFD) to compare the reference clock and the output enable signal to control the output voltage V_{CP} of the charge pump (CP) circuit. Then, V_{CP} is utilized as the gate signal to control the supply voltage of the VCO and also the step delay t_d. The simulation results of the designed PLL are shown in Fig. 14. With the loop established, V_{CP} stabilizes gradually, and the frequency of the VCO stabilizes at 50 MHz indicating the achievement of a 100-ps step delay time t_d. Besides, the designed structure enables the simultaneous generation of signals $V_{E1}\langle 0{:}N-1\rangle$ with varying precisions.

Following the generation of the desired enable signal, we incorporate a signal processing chain to handle the generated signal. Initially, a multiplexer (MUX) is inserted to select a desired overlapping time from $V_{E1}\langle 0{:}N-1\rangle$. In addition, an identical MUX is also introduced in the V_{E2} path to offset the path delay of the MUX. Subsequently, counter-logic blocks

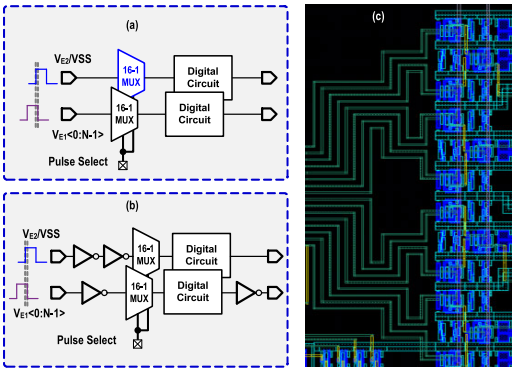


Fig. 15. (a) Redundant design for same latency. (b) Inverted signal for the same rising/falling edge delay. (c) Same path length for the same metal transmission delay.

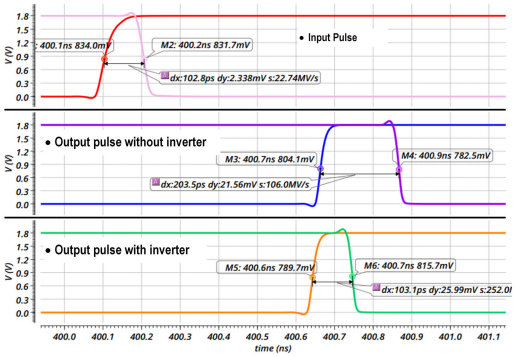


Fig. 16. Simulated results of input pulse and output pulse without and with the additional inverter.

are used to determine the pulse repetition frequency (PRF) of the laser pulse. Next, voltage-controlled delay logic (VCDL) blocks calibrate the delay mismatch during signal propagation. Finally, two level shifters are employed to cope with the ground voltage ringing due to the parasitic inductance.

During the transmission of these two enable signals, three techniques are utilized in this work to reduce the signal processing mismatch, as illustrated in Fig. 15.

1) Redundant design to make sure the same latency across the signal transmission path as shown in Fig. 15(a). Although V_{E2} does not need to be selected by a MUX, a 16–1 MUX is still inserted in the signal transmission path to match the delay of V_{E1}.
2) Due to the fact that the rising edge is the key information utilized for V_{E2}, while the falling edge is crucial for V_{E1}, and considering that the logic circuit exhibits different delays between rising and falling edges, an inverter is inserted in the V_{E2} transmission path, as depicted in Fig. 15(b). This ensures that the MUX and other digital circuits experience the same delay for both V_{E1} and V_{E2} signals. As depicted in Fig. 16, with an input pulse of 102.8 ps, the output pulse deviates to 203.5 ps without adding an inverter in the transmission line, which is entirely unacceptable. However, incorporating the inverter into the transmission line reduces the output pulsewidth to just 103.1 ps, slightly larger than the input signal.
3) The layout is also optimized for uniform length to ensure consistent delay time, as depicted in Fig. 15(c). The serpentine routing technique is employed to achieve this uniform length.

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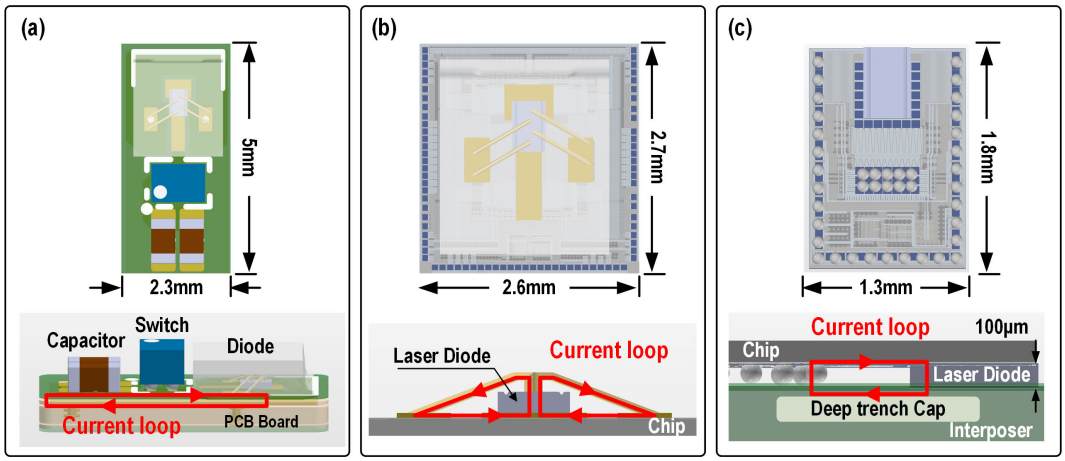


Fig. 17. Three package solutions. (a) Conventional diode-on-PCB. (b) Proposed diode-on-chip. (c) Proposed diode-on-interposer solutions.

B. Low Parasitic Inductance Package Implementation

As illustrated in Fig. 8(c) and (d), parasitic inductance L_P significantly influences the falling edge time and magnitude of the laser diode current. Besides, minimizing the parasitic inductance requires careful consideration of the package implementation during the design phase. As shown in Fig. 17(a), the traditional diode-on-PCB solution places all the capacitor, the switch, and the diode on the PCB. Although some works position the return path directly underneath the top layer's power loop for a smaller physical loop size [11], the parasitic inductance remains large.

In this work, we implement a diode-on-chip solution to directly place a packaged laser diode on chip for a more compact design [28], [29], [30], [31]. As presented in Fig. 17(b), this solution integrates the switch and the bus capacitor into one chip. Due to the low capacitance density of MOM capacitors, only 1.7-nF MOM capacitor is integrated, including M_1–M_5.

Besides, to prevent excessive V_{BUS} voltage drop, we additionally attached two low-inductance silicon capacitors externally in parallel with the ON-chip MOM capacitor. The pads of this configuration are designed specially to accommodate the footprint of the laser diode. In this way, we can eliminate the parasitic inductance of the PCB, with only the bonding wire responsible for the largest part of the parasitic inductance.

Fig. 18 shows the implementation of the proposed diode-on-chip solution. The whole solution only occupies an area of 3.6 × 4.2 mm. The package diode SPL S1L90A_3 used here is from the company Osram. The whole chip is fabricated in a 0.18-μm BCD process and occupies a chip area of 2.7 × 2.6 mm, with the red dashed box representing the pads used to glue or solder the laser diode. Besides, the MOM CAP occupies the largest chip area due to the lower HV capacitance density.

To further mitigate the bonding wire effects, we also design another advanced diode-on-interposer package solution for edge-emitting diodes [32]. This method is not suitable for VCSEL diodes, as the interposer would block vertical light emission. From Fig. 17(c), the laser diode in this solution is directly glued on the interposer and the deep-trench capacitor can be integrated into the interposer [33]. Then, a flipped integrated chip is placed on the top of the laser diode

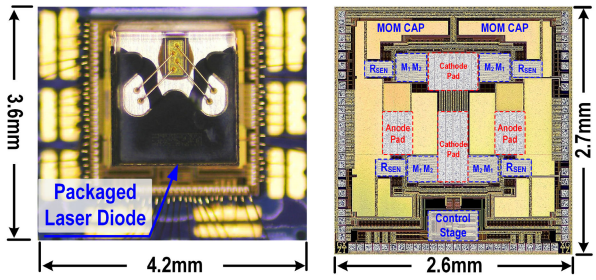


Fig. 18. Diode-on-chip implementation.

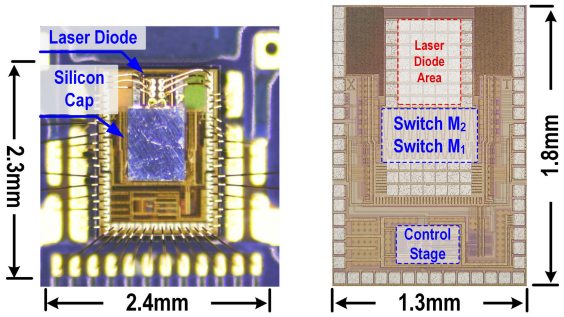


Fig. 19. Diode-on-interposer implementation.

and interposer to form the shortest current loop. In this way, we would eliminate almost all the parasitic packaging inductances except the inherent parasitic inductances of the components.

Fig. 19 shows the implementation of the proposed diode-on-interposer solution. The whole solution just occupies an area of 2.4 × 2.3 mm. The laser diode used in this solution is SPL DS90A_3 from Osram. To emulate the interposer, a silicon capacitor manufactured by Murata is employed here. This high-voltage capacitor, rated at 30 V and 22 nF, features only 20-pH parasitic inductance and a package size of 0201. Besides, it should be noted that the 20-pH parasitic inductance (small-signal model) is only for reference and cannot be directly extrapolated to high-amplitude surge currents. The chip of this solution is also fabricated in 0.18-μm

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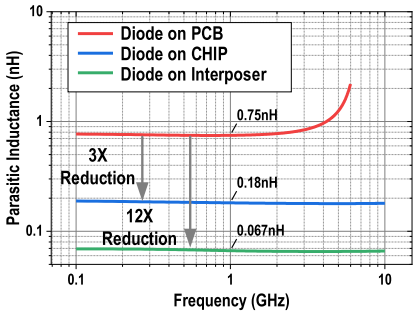


Fig. 20. Simulated parasitic inductance using Ansys HFSS software.

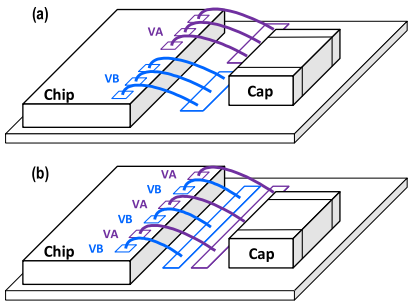


Fig. 21. (a) Conventional and (b) proposed bonding methods.

BCD and occupies a chip area of 1.3 × 1.8 mm. The red dashed box represents the pads used to glue the laser diode. Fig. 20 plots the Ansys HFSS simulation results for the loop parasitic inductance L_P of three different package solutions. (The parasitic inductance of the devices themselves is not included.) In the diode-on-PCB solution, L_P remains at 0.75 nH at 1 GHz. By eliminating the parasitic inductance associated with the PCB, L_P of the diode-on-chip configuration drops to 0.18 nH at 1 GHz. Moreover, by eliminating the bonding wire, the diode-on-interposer solution reduces L_P even further to 0.067 nH at 1 GHz.

C. Low Parasitic Inductance Bonding Wires

Although the previously mentioned diode-on-interposer solution eliminates all bonding wires, its fabrication cost is notable high due to the required interposer. Consequently, bonding wires remain a cost-effective way to connect the chip and PCB board. However, owing to the slender nature of bonding wire, its parasitic inductance is substantial, typically around 1 nH/mm. In this study, due to the constrained chip size for high-voltage MOM capacitors, there is limited capacitance, necessitating the addition of capacitors in parallel outside the chip. Therefore, how to reduce the parasitic inductance of bonding wire is also another issue to be resolved.

Fig. 21(a) illustrates the conventional bonding method, commonly employing multiple bonding wires to minimize the parasitic inductance. Fig. 22 presents the equivalent circuit model of multiple paralleled current loops.

According to the magnetic theory [34], the loop inductance L can be expressed as

$$L = \frac{2E_M}{I^2} = \frac{\sum_{i=1}^{N} L_i I_i^2 + 2\sum_{i=j+1}^{N}\sum_{j=1}^{N} M_{ij} I_i I_j}{\left(\sum_{i=1}^{N} I_i\right)^2} \tag{9}$$

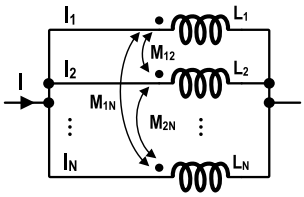


Fig. 22. Equivalent circuit model of multiple paralleled current loops.

where E_M represents the magnetic energy stored in space, encompassing both the self-magnetic energy within each loop and the mutual magnetic energy between the current loops. I_i is the current of the ith loop, L_i is the self-inductance of the ith loop, and M_{ij} is the mutual inductance between the ith and jth loops.

According to (9), the multiple loop parasitic inductance L is primarily determined by two parameters: the current distribution ratio and the mutual inductance M_{ij}. Assume that each loop has an inductance of L_0, the current is evenly distributed into N loops and the mutual inductance M_{ij} is 0. The loop inductance can be reduced to

$$L = \frac{\sum_{i=1}^{N} L_0 I_i^2}{\left(\sum_{i=1}^{N} I_i\right)^2} = \frac{L_0}{N} \tag{10}$$

where N represents how many pairs of bonding wires are being used. However, if all the current flows into one branch, no matter how small the mutual inductance M_{ij} is, the loop inductance L will be increased to

$$L = \frac{L_0 I_1^2}{I_1^2} = L_0. \tag{11}$$

On the other hand, with an even distribution of current flows into each branch, if the current loop is fully coupled ($M_{ij} = L_0$), the loop inductance L remains significant

$$L = \frac{\sum_{i=1}^{N} L_0 I_i^2 + 2\sum_{i=j+1}^{N}\sum_{j=1}^{N} L_0 I_i I_j}{\left(\sum_{i=1}^{N} I_i\right)^2} = L_0. \tag{12}$$

Therefore, to minimize the parasitic inductance, the current needs to be evenly distributed into N branches and the mutual inductance of these branches should be small enough.

In this article, an interleaved bonding method is designed, as demonstrated in Fig. 21(b), featuring an interleaved pad placement and bonding connection. Fig. 23 shows the simulated current density at 1 GHz with $N = 5$ using Maxwell 3-D software. For the conventional bonding method, due to the skin effect of high-frequency current, the current mainly concentrates in the inner loop. Therefore, increasing the number of loops will not significantly change the current distribution. In contrast, the proposed bonding method evenly distributes the current into five loops. Fig. 24 shows the current loop direction of two bonding methods. In the conventional bonding method, the current loops are strongly coupled, while the current loops are reversely coupled in the presented interleaved bonding method.

Fig. 25 presents the simulated loop inductance of two methods, where the bonding wire has a length of 0.75 mm and a height of 0.3 mm. For simulation purposes, the chip and

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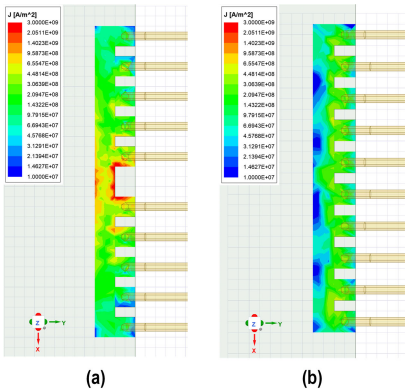


Fig. 23. Simulated current density at 1 GHz with $N = 5$ of (a) conventional and (b) proposed bonding methods.

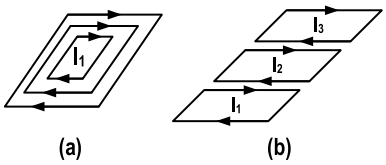


Fig. 24. (a) Current loops are strongly coupled in the conventional method. (b) Current loops are reversely coupled in the proposed method.

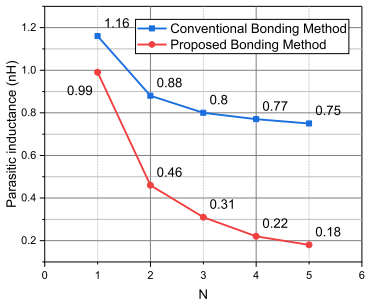


Fig. 25. Simulated parasitic inductance of two bonding methods.

capacitor are treated as ideal conductors. Due to the evenly distributed current and reversely coupled current loop, the inductance of interleaved method drops substantially, while the inductance of conventional method drops just slightly with the increased number of N.

IV. Measurement Results

Fig. 26 presents the optical signal measurement setup of the proposed diode on interposer LiDAR TX. The proposed LiDAR TX board emits the laser pulse and was later received by an optical receiver UPD-35-IR2-P from ALPHALAS. The used optical receiver is an Ultrafast InGaAs p-i-n photodetector with bandwidth >10 GHz. Besides, a switchboard is utilized to select the desired pulsewidth and PRF.

Fig. 27(a) shows the measured control signal V_{G1} and V_{G2} and the received optical pulse signal based on diode on interposer setup. The LiDAR TX works at a maximum PRF of 30 MHz. Fig. 27(b) shows the zoomed-in waveforms. The measured optical signal has 200-ps FWHM, and this is the same as the overlap time of the two control signals V_{G1} and V_{G2}. Besides, there is a delay of 3 ns in the received optical pulse when compared with the control signal because of the system delay and light transmission time. The delay time is

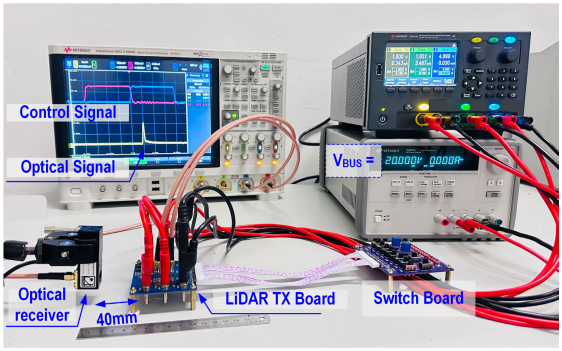


Fig. 26. Measurement setup for optical signal test.

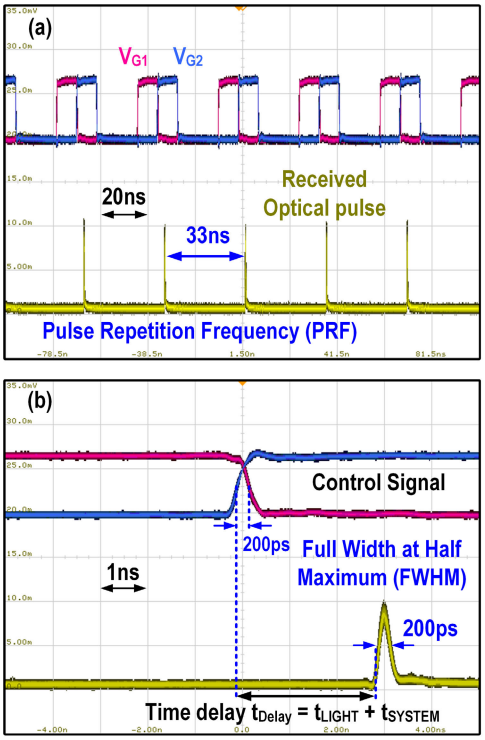


Fig. 27. (a) Measured control signal V_{G1} and V_{G2} and received optical signal at 30-MHz PRF. (b) Zoomed-in waveforms with 200-ps FWHM.

exactly the information used to measure the distance for the LiDAR.

To determine the emitted minimum FWHM, here we employ an oscilloscope with 13.6-GHz bandwidth to measure the received optical signal of the diode on interposer setup. In Fig. 28, we can observe that we obtained a minimum FWHM of 100 ps. Besides, its color grade view is also shown, and the generated optical pulse is stable between cycles. Fig. 29(a) shows the tunable flexibility measurement results of the same setup. The persistence effect feature of oscilloscope is utilized here to enhance the visualization of the laser optical signal over a predetermined time. The persistence effect allows the waveform to remain visible on the screen for a short duration after it is initially displayed. This feature enables us to observe the stability of the received optical signal and determine if it remains consistent over different periods. The FWHM is tuned from 200 ps to 2 ns by shifting the control signal V_{G1}. Besides, the received optical peak magnitude is

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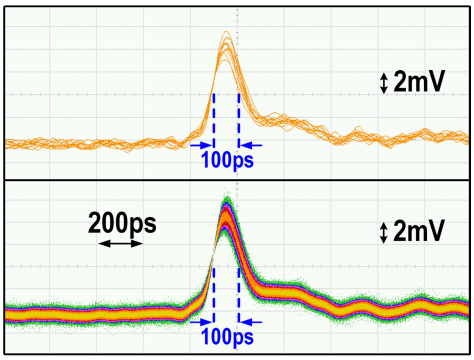


Fig. 28. Measured minimum FWHM of 100 ps using 13.6-GHz bandwidth oscilloscope.

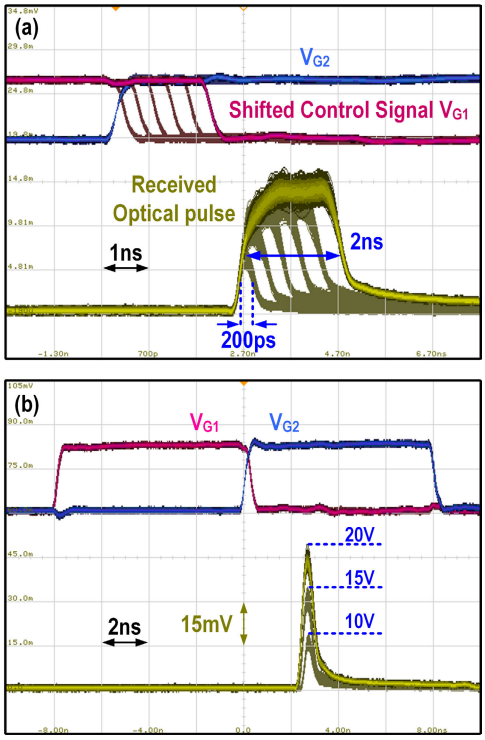


Fig. 29. Measured tunable flexibility of the proposed TX module, with (a) tunable pulsewidth from 200 ps to 2 ns and (b) tunable pulse amplitude.

tuned from 20 to 50 mV by increasing the V_{BUS} voltage from 10 to 20 V, as presented in Fig. 29(b).

In this work, accurately measuring high-speed currents is challenging. Since every resistor possesses a certain inductance, this can immediately distort the results when trying to deduce the current based on the voltage drop across the resistor. Therefore, in this work, we measure the average optical power and indirectly evaluate the resulting current pulse magnitude. Fig. 30 shows the measurement setup for average optical power, with the laser power meter VLP-2000-200 mW carefully positioned to maximize the energy capture. The power meter has a spectral response range of 11–19 000 nm and a resolution of 0.01 mW. In this work, we approximate the received energy P_{RX} to the emitted energy P_{TX} that is

$$P_{\mathrm{TX}} \approx P_{\mathrm{RX}}. \tag{13}$$

As presented in Table III, the received average optical power of the two packages setup is recorded as a function of FWHM

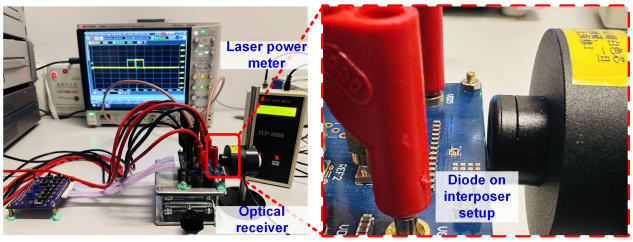


Fig. 30. Measurement setup for average optical power test.

TABLE III
MEASURED RECEIVED AVERAGE OPTICAL POWER

FWHM(ps)	Received optical Power (mW) @T=320ns					
	Diode on Interposer Setup			Diode on Chip Setup		
	VBUS=12	VBUS=15	VBUS=20	VBUS=12	VBUS=15	VBUS=20
100	4.2	6.9	8.6	6.9	12.0	19.5
200	10.2	14.3	18.1	13.0	21.3	33.4
300	17.2	22.6	27.7	19.1	31.2	47.2
400	26.0	34.0	39.2	25.8	40.8	62.9
500	33.6	44.0	51.0	33.2	52.6	79.6
600	42.4	55.4	63.7	39.5	62.4	93.6
700	50.6	66.5	77.2	47.1	73.2	109.0
800	59.4	77.4	90.0	53.0	83.7	124.2
900	68.0	90.0	103.6	59.6	94.1	138.1
1000	76.6	101.3	116.3	66.6	106.8	156.2

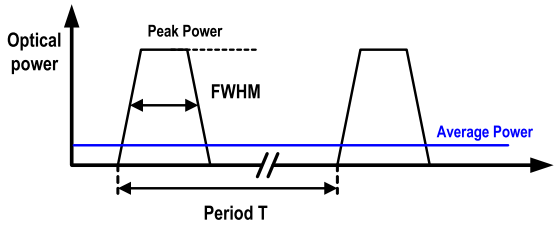


Fig. 31. Relationship between the peak optical power and the average optical power.

TABLE IV
CHIP TEMPERATURE AND ASSOCIATED OPERATING CONDITIONS

FWHM	Frequency	VBUS	Total Power	Emitted power*	Chip Temperature
1ns	6M	12V	0.8W	0.15W	53℃
1ns	12M	12V	1.5W	0.31W	80℃
1ns	3M	20V	1.16W	0.12W	70.5℃
1ns	6M	20V	2.1W	0.24W	125.9℃

*The optical receiver is placed close enough to capture all emitted energy.

over a normalized period of 320 ns and varying the V_{BUS} voltage of 12, 15, and 20 V. According to the relationship shown in Fig. 31, the peak optical power can be calculated using the equation

$$P_{\mathrm{PEAK}} = P_{\mathrm{AVG}} \frac{T}{\mathrm{FWHM}}. \tag{14}$$

As illustrated in Fig. 32, the achieved peak optical power for diode on interposer implementation is 37.2 W with an FWHM of 1 ns and V_{BUS} of 20 V. According to the datasheet of the laser diode SPL S1L90A_3, the achieved current amplitude is approximately 12 A. Besides, we can see with the increase of the pulsewidth, and the peak optical power increases and gradually stabilizes. This phenomenon occurs because shorter widths are significantly influenced by parasitic inductance and resistance, whereas wider pulses primarily depend on parasitic resistance.

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TABLE V
COMPARISON WITH STATE-OF-THE-ART WORKS

	ISSCC'19 [15]	ISPSD'23 [12]	JSSC'23 [4]	EPC9144 [35]	EPC9126 [25]	This Work	
Driver Process/Type	CMOS 55nm	250nm GaN-on-Silicon	180nm BCD	LMG1020	LMG1020	**CMOS 180nm BCD**	
Switch Type	GaN(EPC2040)	Integrated GaN	Integrated LDMOS	GaN (EPC2216)	GaN (EPC2212)	**Integrated LDMOS**	
Laser Diode Type	NA	Resistor	VCSEL*	Edge-Emitting	Edge-Emitting	**Edge-Emitting**	
Laser Diode Driver Type	FET Controlled	FET Controlled	FET Controlled	FET Controlled	Capacitive Discharge	**Double-pulse Overlapping**	
Package solution	Diode on PCB	Diode on PCB	Diode on Chip	Diode on PCB	Diode on PCB	**Diode on Chip**	**Diode on Interposer**
Bus Voltage	10V	15V	12.5V	15V	100V	**20V**	
Pulse Repetition Frequency (PRF)	200M	20M	10M	N/A	N/A	**30MHz**	
Peak Current	28A	10A	8.5A	28A	75A	**15A**	**12A**
Implementation Area**	9.2mm^2	7.1mm^2	13.5mm^2	9.4mm^2	20mm^2	**7.4mm^2**	**2.3mm^2**
Peak Current Density	3.0A/mm^2	1.4A/mm^2	0.6A/mm^2	3.0A/mm^2	3.8A/mm^2	**2.0A/mm^2**	**5.2A/mm^2**
Minumum FWHM	0.9ns	2ns	0.36ns	1.2ns	3ns	**0.1ns**	

* Vertical-Cavity Surface-Emitting Laser, ** Estimated value according to paper or datasheet (Diode, Switch, Capacitor)

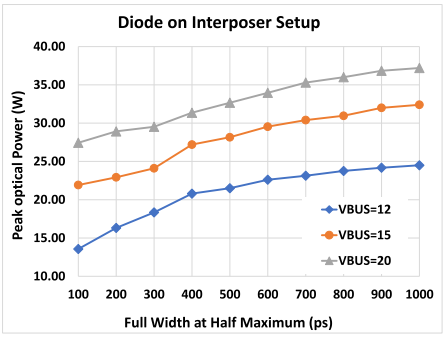


Fig. 32. Calculated peak optical power of the diode on interposer setup versus pulsewidth FWHM.

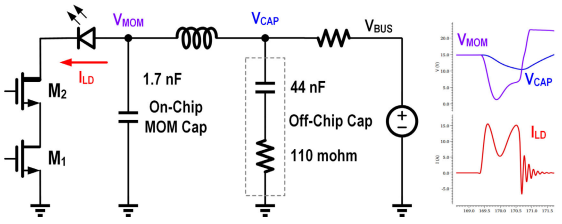


Fig. 33. Simplified circuit model of diode-on-chip solution and simulated waveforms.

However, for diode-on-chip implementation, since the incorporation of low-inductance ON-chip MOM capacitor, the achieved laser diode current is different. Fig. 33 shows a simplified circuit model and simulation results of the laser diode current in this test setup. The MOM capacitor's fully integrated nature results in significantly lower parasitic inductance and resistance, enabling the provision of large peak optical power within very short pulses. However, due to its low capacitance, its voltage V_{MOM} will drop quickly until the OFF-chip capacitors provide the main power. Therefore, (14) may not be fully applicable to this situation anymore, but we

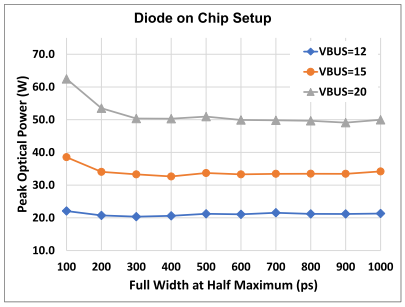


Fig. 34. Calculated peak optical power of the diode-on-chip setup versus pulsewidth FWHM.

can still calculate the achieved peak optical power to compare with another test setup, as shown in Fig. 34. The calculated peak optical power for this implementation is 50 W with an FWHM of 1 ns and V_{BUS} of 20 V. Similarly, the achieved current amplitude is estimated as 15 A. Since the switch size in this setup is twice that of diode-on-interposer implementation, the higher peak optical power within 1 ns is achieved. Besides, we can see with the increase of the pulsewidth, and the peak optical power decreases and gradually stabilizes, which can also verify the simulation and analysis before. The MOM capacitor provides a large peak optical current within very short pulsewidth.

In addition, although Fig. 27 shows the maximum PRF of 30 MHz and Fig. 32 shows the maximum peak power of 37.2 W, the maximum PRF and maximum output power cannot be achieved simultaneously due to excessive heat on the chip. As depicted in Fig. 35, thermal images with fan cooling are captured with 1-ns FWHM at varying V_{BUS} and pulse repetition frequencies. With increased frequency and higher bus voltage, the chip becomes excessively hot and reaches its thermal limit. Table IV summarizes the chip temperature and associated operating conditions. With the TX operating at an

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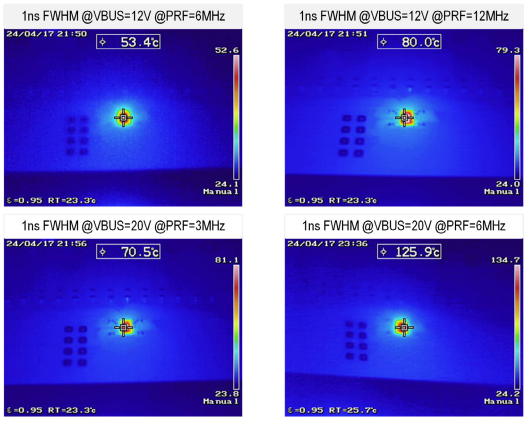


Fig. 35. Thermal images with fan cooling at 1-ns FWHM and varying bus voltage and frequency.

FWHM of 1 ns, a PRF frequency of 6 MHz, and V_{BUS} voltage of 20 V, the chip temperature reaches its limit to 126 °C.

In Table V, we summarized the performance of the proposed laser driver and compared it with state-of-the-art designs [35]. To our knowledge, this is the first work that utilizes the DPO time to generate an ultra-short laser pulse of 100 ps–2 ns. Besides, we also implement two packaging solutions to obtain a high peak output current. With the proposed structure and implemented package solution, the detecting resolution and range of the LiDAR system can be greatly improved. Thanks to the designed package solutions, the TX systems achieve a peak current density of 2.0 and 5.2 A/mm^2 within 1-ns pulsewidth. For the diode-on-chip setup, the implementation area also includes two OFF-chip silicon capacitors.

V. Conclusion

The detection resolution and range of dToF LiDAR systems depend on the minimum width and maximum power of the emitted laser pulse. In this work, we proposed a DPO scheme to eliminate the limitation of conventional pre-drivers. Therefore, the generated laser pulsewidth can be extremely shorter than the control signal. We also implement two package solutions to reduce parasitic inductance and obtain high optical power for laser pulses. In this way, we can obtain a high-resolution and long-range LiDAR TX system. With the proposed structure and package solutions fabricated in a 0.18-μm BCD process, the two implemented TX systems achieve a minimum pulsewidth of 100 ps or a maximum output current of 15 A within 1-ns pulsewidth.

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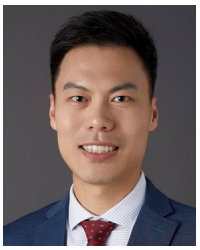
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